

PAVAN KOUNDINYA LANKA

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Professional Summary

Software Development Engineer (SDE-2) with expertise in geospatial platforms, satellite ground systems, imagery tasking workflows, and scalable backend systems. Experienced in building STAC-compliant data dissemination platforms, mission control applications, geospatial automation pipelines, and cloud-native services using FastAPI and AWS. Strong background in remote sensing systems, geospatial visualization, and operational tooling for multi-constellation satellite operations.

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Technical Skills

Programming Languages: Python, JavaScript, Bash, HTML5, CSS3

Backend Development: FastAPI, Django, RESTful API Development, JWT Authentication, API Design, Uvicorn

Frontend Development: React.js, Redux, JavaScript (ES6+), HTML5, Bootstrap, Tailwind CSS

Geospatial & Remote Sensing: GDAL, RasterIO, STAC, GeoTIFF, Cloud Optimized GeoTIFF (COG), GeoJSON, Satellite Imagery Processing

Cloud & DevOps: AWS (EC2, S3, RDS, VPC), Docker, Apache, Linux, Git, Shell Scripting, Cron Jobs

Data Engineering & ML: NumPy, Pandas, OpenCV, Scikit-learn, Image Processing, Data Automation Pipelines

Databases: PostgreSQL

Tools & Platforms: Postman, CVAT, Mission Control Systems, Geospatial Data Workflows

Domain Expertise: Satellite Ground Systems, Imagery Tasking Platforms, Geospatial Data Dissemination, Conflict Management Systems, Remote Sensing Analytics “

Professional Experience

Software Development Engineer 2

Oct 2025 – Present

Kaleideo Space Systems

Bangalore, India

- Developed a STAC-compliant Data Dissemination Platform enabling standardized discovery and access to multi-source satellite imagery datasets including Landsat, Sentinel, and Cartosat.
- Built an end-to-end Imagery Tasking Platform supporting order management, scheduling, processing, and delivery workflows across multiple satellite constellations.
- Engineered a CVAT automation pipeline for large-scale geospatial labeling workflows supporting GeoTIFF, COG, and TIFF imagery formats with automated PNG conversion and GeoJSON annotation export.
- Designed and developed a Mission Control application with real-time conflict visualization and scheduling management capabilities for constellation operations.
- Built scalable backend services using FastAPI and PostgreSQL with secure REST APIs supporting high-performance geospatial operations.
- Developed interactive frontend interfaces for operational monitoring, task management, and geospatial visualization workflows.

Software Engineer

Jan 2023 – Oct 2025

Azista Industries Pvt Ltd

India

- Contributed to the development and deployment of satellite ground system applications including mission control, health monitoring, data dissemination, and live satellite tracking systems.
- Improved operational efficiency through workflow automation using shell scripting, cron jobs, and AWS-based deployments on EC2 infrastructure.
- Developed secure backend services using FastAPI and JWT authentication for mission-critical operational systems.
- Built responsive frontend interfaces and interactive visualization modules for telemetry monitoring and operational data exploration.
- Implemented real-time satellite tracking and visualization features to enhance mission situational awareness.
- Applied image processing techniques including mosaicking, geometric correction, and band registration for satellite imagery workflows.

Transaction Risk Investigator

Sep 2021 – Sep 2022

Amazon

India

- Worked on transactional risk analysis and fraud detection workflows leveraging customer behavior and transaction pattern analysis.
- Conducted investigations on suspicious activities across global marketplaces to support fraud prevention and loss mitigation.
- Collaborated with cross-functional teams to improve fraud detection methodologies and operational processes.

Process Executive

Oct 2019 – Jul 2021

Cognizant

India

- Managed mapping operations and geospatial data updates across 26 countries for Google Maps platforms.
- Performed aerial imagery analysis and map validation workflows using Google’s internal ATLAS platform.
- Modeled road infrastructure updates and validated geographic datasets to maintain mapping accuracy and consistency.

Projects

IoT-Based Smart Garbage Alert System

- Developed an IoT-enabled waste monitoring system using ultrasonic sensors and NodeMCU to automate garbage collection alerts and improve waste management efficiency.

Education

KL University

Bachelor of Technology in Computer Science Engineering

2018 – 2022

CGPA: 6.74